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THIQA CAR INSURANCE

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Certificate

It is certified that this project report has been prepared and written under my direct supervision and guidance. The project report is approved for submission for its evaluation.

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D e d i c a t i o n

We dedicate this work to our families, whose continuous support and encouragement made this achievement possible. We also dedicate it to our teachers and mentors at the Information Technology Department, Qassim University, for their guidance throughout our studies.

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The Project Team

A c k n o w l e d g e m e n t

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The Project Team

A b s t r a c t

Choosing a suitable car insurance policy has become increasingly difficult in today's digital environment. Most existing platforms provide general quotations and static policy information without enough personalized guidance, which makes it hard for drivers to compare coverage options, understand exclusions and deductibles, and select a policy that matches their vehicle and driving profile. This project proposes an intelligent car insurance recommendation system, THIQA, that focuses only on motor insurance and uses driver-specific and vehicle-specific data to deliver personalized, data-driven recommendations.

The system collects information such as driver age, vehicle type, vehicle value, expected mileage, driving frequency, accident history, and preferred coverage level, then processes it through a set of collaborative agents responsible for driver profiling, vehicle risk evaluation, coverage matching, recommendation, and explanation. This multi-agent approach aims to simplify policy selection, increase transparency by explaining each recommendation, and improve the overall digital customer experience.

This report presents the problem definition, objectives, scope, and study plan, followed by a literature review of e-commerce in car insurance, motor insurance concepts, auto InsurTech, digital transformation, the Saudi car insurance ecosystem, and the proposed solution.

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List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
B.Sc.	Bachelor of Science
FSDP	Financial Sector Development Program
InsurTech	Insurance Technology
IoT	Internet of Things
KSA	Kingdom of Saudi Arabia
ML	Machine Learning
OECD	Organisation for Economic Co-operation and Development
SAMA	Saudi Central Bank
UBI	Usage-Based Insurance
UNCTAD	United Nations Conference on Trade and Development
UML	Unified Modeling Language
MAS	Multi-Agent System
JSON	JavaScript Object Notation
XAI	Explainable Artificial Intelligence
REST	Representational State Transfer
TPL	Third-Party Liability
ERD	Entity Relationship Diagram
PK	Primary Key
FK	Foreign Key

CHAPTER ONE: INTRODUCTION

1.1 Problem Specification and Motivation

Most existing car insurance platforms provide general quotations and static policy information without offering enough personalized guidance. Drivers and vehicle owners often need to compare many policy options, coverage limits, exclusions, deductibles, and prices. This makes the decision-making process confusing, especially when users do not clearly understand the difference between third-party liability insurance, comprehensive insurance, and optional add-ons.

As a result, users may rely on incomplete information when choosing car insurance. A driver may select a policy that does not match the vehicle value, driving behavior, claim history, or financial ability. In other cases, the user may overpay for coverage that is not necessary, or may buy the cheapest option without understanding the risks of limited protection.

The motivation behind this project is to address these challenges by developing an intelligent car insurance recommendation system. The proposed system focuses only on car insurance and uses user-specific and vehicle-specific data to provide personalized, data-driven recommendations. By applying a multi-agent approach, the system aims to simplify policy selection, improve transparency, and enhance the overall customer experience in digital motor insurance services.

1.2 Goals and Objectives

The main goal of this project is to design and analyze an intelligent car insurance recommendation system that helps users select the most suitable motor insurance policy based on driver profile, vehicle information, and insurance needs.

The specific objectives of the project are as follows:

- To design a system that collects and processes driver-specific and vehicle-specific data.
- To analyze factors such as driver age, vehicle type, vehicle value, driving frequency, annual mileage, accident history, and preferred coverage level.
- To develop a multi-agent decision-making approach for car insurance recommendation.
- To recommend suitable car insurance options such as third-party liability insurance, comprehensive insurance, and usage-based insurance.
- To enhance transparency by explaining why a specific car insurance policy is recommended.
- To improve the digital customer experience by reducing confusion during online car insurance selection.

1.3 Project Scope

This project focuses on the design and implementation of an intelligent car insurance recommendation platform. The system is based on user-provided driver and vehicle information, such as age, vehicle details, vehicle value, purpose of use, coverage preferences, and other data commonly required by motor insurance providers.

The scope is limited to car insurance only. It does not include health insurance, life insurance, travel insurance, property insurance, or other non-motor insurance products. The system is designed to help users compare available motor insurance offers and receive a recommendation for the most suitable insurance package based on their profile and preferences.

The proposed solution follows a multi-agent architecture consisting of three cooperating agents. The Data Receiver and Retrieval Agent collects customer information and retrieves available insurance offers from connected providers. The Recommendation Agent evaluates and ranks the retrieved offers using predefined recommendation criteria. The Explainer Agent generates a clear justification describing why the selected package was recommended.

The platform retrieves insurance offers from providers, evaluates available options, and generates personalized and explainable recommendations to support users in selecting the most suitable insurance package.

1.3.1 Guiding Principles

The development of this study is guided by the following principles:

- **User-Centric Design:** Ensuring that the system is easy for drivers and vehicle owners to use.
- **Car Insurance Focus:** Ensuring that all recommendations are limited to motor insurance products and vehicle-related risks.
- **Data-Driven Decision Making:** Using driver and vehicle data to generate more accurate car insurance recommendations.
- **Transparency:** Providing clear explanations for each recommendation, including the main risk and coverage factors considered.
- **Privacy and Security:** Protecting sensitive user data such as personal details, vehicle information, and claim history.
- **Scalability:** Designing the system so that future versions can include telematics, real-time driving data, and integration with digital insurance platforms.

1.3.2 Target Audience

The intended audience of this project includes:

- Drivers and vehicle owners seeking suitable car insurance coverage.
- New drivers who need guidance in understanding motor insurance options.
- Users comparing third-party liability and comprehensive car insurance.
- Car insurance companies aiming to improve digital customer engagement.
- Insurance technology providers developing intelligent motor insurance services.

1.4 Project Plan and Schedule

This section outlines the project timeline and key phases to ensure structured and efficient progress. The schedule focuses on research, analysis, system design, and reporting for a car insurance recommendation system.

Task	Focus Area	Expected Date
Project Planning and Topic Approval	Confirm car insurance as the only project domain	Feb 15
Literature Review and Research	Review car insurance, auto InsurTech, and digital motor insurance platforms	Mar 8
E-Commerce and Car Insurance Sections	Study online policy quotation, purchase, renewal, and digital payments	Mar 22
Technology and InsurTech Sections	Analyze AI, telematics, IoT, automation, and claims technologies for car insurance	Apr 5
Market Analysis	Analyze global and Saudi car insurance digital platforms and market needs	Apr 19
Problem Identification	Define user difficulties in choosing suitable car insurance policies	Apr 26
Proposed Solution Design	Design the multi-agent car insurance recommendation system	May 3
Report Writing and Integration	Write and revise the full report chapters	May 10
Editing, Formatting, and Review	Check consistency, citations, formatting, and final structure	May 17
Presentation Preparation and Final Submission	Prepare final submission and presentation materials	May 21

Table 1.1: Project Study Plan and Schedule

This section outlines the project timeline and key phases to ensure structured and efficient progress. The Gantt chart below illustrates the distribution of tasks and their expected duration.

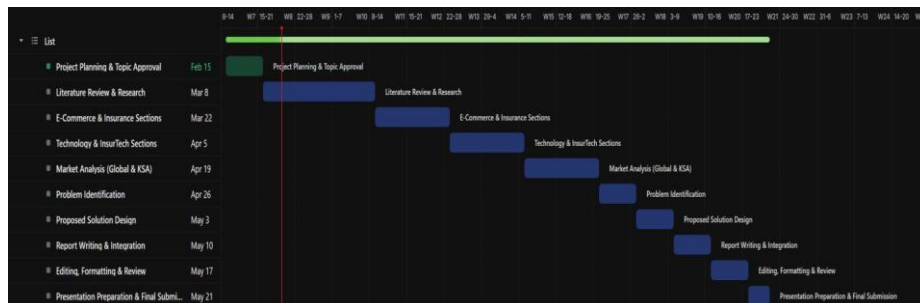


Figure 1.1: Project Gantt Chart (Project Plan and Schedule)

1.5 Organization of the Chapters

This report is organized into three main chapters:

- Chapter 1: Introduction – This chapter introduces the car insurance recommendation project, defines the problem, outlines the objectives, and presents the study scope and plan.
- Chapter 2: Literature Review – This chapter provides an overview of e-commerce in car insurance, motor insurance concepts, auto InsurTech, telematics, digital transformation, and car insurance market challenges.
- Chapter 3: Proposed Car Insurance Recommendation Solution – This chapter presents the proposed intelligent car insurance recommendation system, including its design, multi-agent approach, and expected benefits.

1.6 Summary

This chapter introduced the main problem of selecting appropriate car insurance in a complex digital environment. It highlighted the limitations of current car insurance platforms, including lack of personalization, unclear explanations, and difficulty comparing coverage options. The chapter also presented the motivation for developing an intelligent car insurance recommendation system and outlined the objectives, scope, guiding principles, target audience, schedule, and overall structure of the report.

CHAPTER TWO: LITERATURE REVIEW

2.1 E-Commerce in Car Insurance

2.1.1 Definition of E-Commerce

Electronic Commerce (E-Commerce) refers to the process of buying, selling, transferring, or exchanging products, services, and information through computer networks, mainly the Internet. It includes digital business activities such as online marketing, online service delivery, electronic payments, and customer support [1].

In the context of car insurance, e-commerce allows drivers to compare prices, request quotations, buy policies, renew coverage, submit documents, pay premiums, and access policy information through websites or mobile applications. This digital process reduces the need for physical branch visits and supports faster customer service.

Digital payments are especially important for online car insurance because policy issuance usually depends on successful payment confirmation. Secure payment channels such as credit cards, debit cards, e-wallets, and bank transfers help insurers issue motor insurance policies quickly while protecting customer information through encryption and authentication [2].

2.1.2 Relationship Between E-Commerce and Car Insurance

E-commerce and car insurance are closely connected because many motor insurance services can now be completed online. Digital platforms allow users to enter vehicle information, driver details, and coverage preferences, then receive quotations from different insurance providers. This supports faster comparison and enables customers to purchase car insurance from anywhere.

Online car insurance platforms also support embedded insurance models. For example, a customer buying a car from a dealership or online vehicle platform may be offered motor insurance during the purchase process. Similarly, ride-hailing, car rental, vehicle financing, and fleet management platforms may integrate insurance products directly into their digital services.

The relationship between e-commerce and car insurance also extends to claims management. Customers can report accidents, upload vehicle damage photos, track claim status, and receive repair approvals through digital channels. This makes car insurance more convenient and increases the importance of secure, reliable, and user-friendly digital insurance systems.

2.1.3 Risks in Online Car Insurance Transactions

Although online car insurance platforms are convenient, they introduce several risks that must be managed carefully:

- **Cybersecurity Threats:** Attackers may attempt to steal personal data, vehicle information, payment details, or policy documents.
- **Financial Fraud:** Fake transactions, stolen payment cards, identity theft, and false customer information may affect online insurance operations.
- **Data Privacy Risks:** Car insurance platforms process sensitive information such as national identity numbers, driving history, accident records, and vehicle ownership data.

- **Operational Risks:** Website downtime, system errors, failed payment processing, or incorrect quotation logic may affect the customer experience.
- **Digital Misunderstanding:** Users may buy unsuitable coverage if policy limits, exclusions, deductibles, or claim requirements are not clearly explained.

2.2 Car Insurance

2.2.1 Definition of Car Insurance

Car insurance, also known as motor insurance or auto insurance, is a financial protection contract between a vehicle owner and an insurance company. The policy provides protection against specified vehicle-related risks such as traffic accidents, third-party liability, theft, fire, and physical damage to the insured vehicle. In exchange, the policyholder pays a premium to the insurer [3].

The main purpose of car insurance is to transfer the financial burden of vehicle-related losses from the individual driver to the insurer. At the individual level, car insurance protects drivers from unexpected repair costs and legal liabilities. At the wider economic level, it supports road safety, financial stability, and confidence in vehicle ownership and transport activities.

2.2.2 Types of Car Insurance

Car insurance products can be classified into several common types:

1. **Third-Party Liability Insurance:** Covers damages or injuries caused by the insured driver to other people, vehicles, or property. This is usually the minimum required form of motor insurance in many markets.
2. **Comprehensive Car Insurance:** Provides broader protection by covering third-party liability as well as damage to the insured vehicle, depending on policy terms and exclusions.
3. **Collision Coverage:** Covers damage to the insured vehicle resulting from a collision with another vehicle or object.
4. **Theft and Fire Coverage:** Covers losses caused by vehicle theft, attempted theft, or fire, according to policy conditions.
5. **Usage-Based Insurance (UBI):** Uses driving data such as mileage, speed, braking, acceleration, and time of driving to calculate premiums or offer personalized pricing.
6. **Add-On Covers:** Optional benefits such as roadside assistance, agency repair, rental car replacement, windshield repair, or natural disaster coverage.

2.2.3 Traditional Car Insurance Model

The traditional car insurance model relies on risk pooling and historical data. Insurance companies collect premiums from many policyholders and use statistical models to estimate expected losses. Premiums are usually determined by factors such as driver age, vehicle type, vehicle value, claim history, geographic area, and selected coverage level [4].

In traditional motor insurance, underwriting and claims may involve manual document review, vehicle inspection, call center communication, and branch-based customer service. Although this model can be effective, it is often slower and less personalized than modern digital insurance models.

2.2.4 Challenges in Traditional Car Insurance

Traditional car insurance faces several challenges:

- **Limited Personalization:** Premiums may be based on broad customer categories rather than actual driving behavior.
- **Slow Claims Processing:** Accident reporting, vehicle inspection, repair approval, and settlement may require manual steps.
- **Fraudulent Claims:** False accident reports, exaggerated repair costs, staged accidents, and duplicate claims can increase insurer losses.
- **Information Asymmetry:** Drivers may know more about their driving behavior and vehicle usage than the insurer.
- **Customer Confusion:** Policy exclusions, deductibles, coverage limits, and add-ons may be difficult for users to understand.
- **Operational Inefficiency:** Legacy systems and paper-based processes increase cost and reduce service speed.

2.3 Car Insurance Technology (Auto InsurTech)

2.3.1 Definition and Evolution of Auto InsurTech

InsurTech refers to the use of emerging technologies and innovative business models to transform insurance services [14]. In car insurance, Auto InsurTech focuses on improving motor insurance quotation, underwriting, pricing, policy issuance, claims processing, fraud detection, and customer service.

Auto InsurTech uses technologies such as artificial intelligence, mobile applications, telematics, Internet of Things (IoT), big data analytics, cloud computing, computer vision, and application programming interfaces (APIs). These technologies help insurers move from static and manual processes toward faster, data-driven, and personalized car insurance services [15].

2.3.2 Key Drivers of Auto InsurTech Growth

The growth of Auto InsurTech is driven by several factors:

1. **Technology:** Artificial intelligence can analyze large datasets and support risk assessment, fraud detection, and claims automation. Big data can process driving behavior, accident records, and customer profiles. Cloud services support scalability and lower infrastructure cost.
2. **Cost Reduction:** Automation reduces manual work in underwriting, policy issuance, and claims handling. Lower operating costs may improve pricing and service efficiency.

3. **Changing Customer Expectations:** Modern customers expect fast quotations, online policy purchase, digital payment, and real-time claim status tracking.
4. **Competition:** Digital aggregators and InsurTech startups pressure traditional insurers to improve speed, transparency, and customer experience.
5. **Regulatory Support:** Regulatory sandboxes and digital insurance frameworks allow innovative motor insurance solutions to be tested while maintaining consumer protection.

2.3.3 Difference Between Traditional Car Insurance and Auto InsurTech

Traditional car insurance usually relies on manual underwriting, standard pricing categories, physical inspections, and branch-based or call-center service. In contrast, Auto InsurTech uses digital platforms, automation, data analytics, and AI to improve the entire motor insurance value chain.

The main difference lies in the level of technological integration. Auto InsurTech supports instant quotation, online policy issuance, telematics-based pricing, digital claims submission, computer vision damage assessment, automated fraud detection, and personalized recommendations. Traditional models are often slower and less flexible because they depend more heavily on manual processes and historical averages [16].

2.4 Digital Transformation in Car Insurance

Digital transformation in car insurance is the redesign of business models, operational processes, pricing methods, risk assessment, and claims services using digital technologies. It enables insurers to improve efficiency, personalize coverage, reduce fraud, and offer better customer experiences [9].

2.4.1 Automation of Processes and Digital Value Chains

Automation in car insurance relies on digital platforms, APIs, workflow engines, and rule-based systems. These tools allow insurers to issue quotations, verify customer information, calculate premiums, generate policy documents, and process payments with less manual intervention.

Automated underwriting evaluates driver and vehicle data against predefined rules. For example, the system may consider vehicle value, driver age, accident history, policy type, selected deductible, and claim history to determine eligibility and pricing. This reduces processing time and improves consistency.

APIs also allow car insurance services to connect with e-commerce platforms, car dealerships, vehicle financing providers, accident reporting services, and digital payment providers. As a result, customers can receive insurance offers at the point of vehicle purchase or renewal.

2.4.2 Data-Driven Risk Assessment and Artificial Intelligence

Data-driven risk assessment uses structured and unstructured data to understand the risk level of drivers and vehicles. In car insurance, useful data may include vehicle characteristics, mileage, location, claim history, repair history, driving behavior, accident frequency, and telematics signals [11].

Data analytics can be applied at four levels: descriptive analytics to understand past claims, diagnostic analytics to identify causes of losses, predictive analytics to forecast future accident risk, and prescriptive analytics to recommend optimal pricing or coverage decisions [11].

Artificial intelligence and machine learning can improve car insurance by identifying patterns that are difficult to detect manually. AI models can support risk segmentation, premium calculation, fraud detection, customer service chatbots, claim triage, and image-based vehicle damage assessment [10].

2.4.3 Internet of Things, Telematics, and Real-Time Monitoring

The Internet of Things (IoT) refers to physical objects embedded with sensors that transmit data to digital systems [12]. In car insurance, IoT is strongly connected to telematics, where devices or mobile applications collect driving data and send it to insurers for analysis.

Telematics can monitor factors such as annual mileage, speed, harsh braking, rapid acceleration, cornering behavior, time of driving, location patterns, and accident events. This supports usage-based insurance, where the premium can reflect how much and how safely a person drives [17].

Real-time monitoring can also support accident detection, emergency assistance, theft recovery, and preventive risk alerts. However, it requires strong data security, customer consent, privacy protection, and clear explanation of how driving data affects premiums.

2.4.4 Fraud Detection, Blockchain, and Cybersecurity

Car insurance fraud may include staged accidents, false injury claims, inflated repair invoices, duplicate claims, and incorrect driver information. AI-based fraud detection uses pattern recognition, link analysis, and anomaly detection to identify suspicious claims and relationships between claimants, repair centers, and service providers.

Blockchain can support transparency and traceability by creating shared records that are difficult to alter without validation [13]. In car insurance, blockchain may be used for policy verification, claim history tracking, repair documentation, or smart contracts. However, adoption depends on interoperability, regulation, governance, and data standardization.

Cybersecurity is also essential because digital car insurance platforms process sensitive personal, financial, and vehicle data. Strong authentication, encryption, secure APIs, access controls, and regular monitoring are required to protect customers and insurers.

2.4.5 Customer Experience Enhancement and Cost Optimization

Digital transformation improves the customer experience by making car insurance faster and easier to understand. Customers can compare quotations online, select coverage, pay premiums, receive digital policies, renew insurance, and track claims without visiting a branch.

Computer vision can improve claims processing by allowing customers to submit photos of vehicle damage. AI models can estimate repair severity, classify damage type, and support faster claim decisions. This reduces waiting time and improves customer satisfaction.

Cloud computing and automation can reduce capital expenditure and operating costs by replacing legacy systems with scalable digital services. Over time, car insurance can shift from

reactive compensation after accidents toward proactive prevention through driving alerts, safety scoring, and personalized risk advice.

2.5 Car InsurTech in Saudi Arabia

2.5.1 Car Insurance in Saudi Arabia

The Saudi insurance market has been transformed by digital services, regulatory development, and increasing customer demand for online insurance solutions. Motor insurance is one of the most important insurance lines in Saudi Arabia because vehicle ownership, road mobility, and third-party liability protection create strong demand for car insurance services [7].

Saudi car insurance customers increasingly use digital channels to compare prices, purchase policies, renew coverage, and access claim services. This creates opportunities for Auto InsurTech solutions that improve pricing accuracy, transparency, and user experience.

2.5.2 Government Initiatives and Regulations

The regulatory environment in Saudi Arabia supports the development of digital financial and insurance services. Insurance regulation has moved toward a more specialized structure, and digital insurance activities are increasingly expected to meet consumer protection, data privacy, and operational reliability requirements [5].

For car insurance, digital transformation must also consider accident reporting, claim documentation, policy verification, and integration with relevant national platforms. Regulatory sandboxes can support testing of innovative models such as pay-per-mile insurance, AI-based claims assessment, and telematics-based pricing.

2.5.3 Role of Vision 2030

Vision 2030 supports financial sector development, digital transformation, cashless payments, and innovation in Saudi Arabia. These goals encourage insurers and technology providers to build digital services that improve access, efficiency, and customer experience [6].

In the car insurance sector, Vision 2030 creates a favorable environment for mobile-first insurance services, digital comparison platforms, automated policy issuance, and advanced analytics. These developments can help improve insurance penetration, market transparency, and service quality.

2.5.4 Local Car Insurance Digital Ecosystem and Players

The Saudi digital car insurance ecosystem includes comparison platforms, insurance aggregators, traditional insurance companies, technology providers, and service partners such as repair networks and accident support providers. Platforms such as Tameeni, BCare, and similar aggregators allow customers to compare and purchase motor insurance policies online [7].

Traditional insurers are also investing in digital channels to improve quotation, policy management, renewal, claims submission, and customer support. Cooperation between insurers

and technology providers can accelerate Auto InsurTech development by combining insurance expertise with digital innovation.

2.5.5 Insurance Regulatory Sandbox Framework (Insurance Lab)

The insurance regulatory sandbox, also known as an insurance lab, provides a controlled environment for testing innovative insurance business models. In the car insurance sector, this framework can support experiments with telematics, usage-based pricing, AI-driven claims processing, and digital customer onboarding [8].

Such frameworks are important because they allow innovation while maintaining consumer protection and regulatory supervision. They can help insurers evaluate new car insurance technologies before full market deployment.

2.6 Types / Classifications of Car InsurTech

2.6.1 Digital Car Insurance Platforms

Digital car insurance platforms allow users to obtain quotations, compare policies, purchase coverage, renew policies, and manage claims through websites or mobile applications. These platforms reduce the time required to complete insurance transactions and give users more control over policy selection.

They may include comparison tools, online payment, policy document generation, customer support chatbots, claim tracking, and integration with vehicle databases. Their main value is simplifying access to car insurance and increasing transparency.

2.6.2 Usage-Based Insurance

Usage-Based Insurance (UBI), also called telematics insurance, is a car insurance model in which the insurer monitors driving behavior or vehicle usage to determine premiums. Data may be collected through a mobile application, plug-in device, or built-in vehicle system [17].

Common telematics factors include:

- Number of miles driven.
- Time of day when the vehicle is used.
- Driving location or road type.
- Speeding behavior.
- Rapid acceleration.
- Harsh braking.
- Sharp cornering.
- Airbag deployment or accident signals.

UBI supports more personalized pricing because safer drivers may receive better rates, while high-risk behavior can be reflected in premium calculation. However, UBI requires careful privacy management and clear customer consent.

2.6.3 Embedded Car Insurance

Embedded car insurance refers to offering motor insurance inside another digital customer journey. For example, a customer buying a vehicle, financing a car, renting a car, or using a mobility platform may receive an insurance offer at the point of transaction.

This model can reduce friction because users do not need to search separately for insurance. It also allows insurers to reach customers at the exact moment when coverage is needed.

2.6.4 AI-Driven Car Insurance Solutions

AI-driven car insurance solutions use artificial intelligence, machine learning, automation, and data analytics to improve coverage, pricing, underwriting, fraud detection, and claims service [18].

In motor insurance, AI can analyze claim descriptions, driving behavior, images of vehicle damage, repair invoices, and customer profiles. It can also support personalized recommendations by matching user needs with suitable policy types.

2.6.5 Blockchain-Based Car Insurance

Blockchain is a shared and tamper-resistant digital ledger that can record transactions and track assets in a business network [19]. In car insurance, blockchain may support trusted policy verification, claim history records, smart contract execution, and secure data sharing between insurers, repair centers, regulators, and customers.

Blockchain-based car insurance could improve transparency, reduce fraud, and speed up settlements. However, it requires coordination among stakeholders and compatibility between systems.

2.7 Car InsurTech Market Analysis

The InsurTech market is growing as insurance companies adopt cloud systems, automation, AI, telematics, and digital platforms [20]. Within this wider market, car insurance is one of the most suitable areas for digital transformation because vehicles generate valuable data and customers frequently need quotation, renewal, and claims services.

Car InsurTech growth is influenced by several trends:

- Increasing use of online comparison platforms for motor insurance.
- Growth of telematics and usage-based insurance models.
- Rising demand for faster digital claims processing.
- Use of computer vision to estimate vehicle damage.
- Integration of car insurance into vehicle sales, financing, and mobility platforms.
- Need for fraud detection and cost control in motor claims.

2.7.1 Global Car Insurance Technology Companies

Several global InsurTech companies demonstrate how technology can reshape insurance services. Some companies focus on AI-powered customer service and claims automation, while others focus on embedded insurance, fraud detection, or telematics-based pricing [20], [21], [22].

For car insurance, the most relevant global technology patterns include digital quotation engines, mobile-first insurance apps, automated claims workflows, driver scoring systems, vehicle damage image analysis, and API-based partnerships with mobility platforms. These patterns show that the future of car insurance depends on data, automation, personalization, and customer experience.

2.7.2 Problem to Be Solved

2.7.2.1 Key Issues

In today's digital environment, users face difficulty in selecting appropriate car insurance due to the wide variety of coverage options, pricing differences, and lack of policy knowledge. Most platforms provide comparison results but do not always explain which policy is best for a specific driver and vehicle profile.

2.7.2.2 Impact

- On Customers: Users may choose unsuitable car insurance, leading to insufficient coverage, unnecessary costs, or misunderstanding of claim conditions.
- On Insurance Companies: Insurers may struggle to match the right motor insurance product with the right customer, which can reduce satisfaction and increase policy cancellation or complaint rates.
- On the Market: Lack of transparency may reduce customer trust in online car insurance platforms and slow adoption of digital insurance services.

2.7.2.3 Existing Limitations

Current systems often rely on basic filtering by price, company name, and coverage type. They may not provide intelligent, explainable, or personalized recommendations. They also may not consider enough risk factors such as driving behavior, annual mileage, vehicle usage, previous claims, and user coverage preferences. This limits the effectiveness of digital car insurance platforms.

2.8 Proposed Solution

The proposed solution is THIQA, an intelligent car insurance recommendation platform designed to help users identify the most suitable motor insurance package based on their vehicle information, driver profile, and insurance preferences. Instead of manually comparing multiple insurance offers, users provide the information required for obtaining insurance quotations, and the system automatically analyzes the available options and recommends the most suitable package.

The platform is based on a multi-agent architecture consisting of three specialized agents that cooperate to produce the final recommendation.

The first agent, the Data Receiver and Retrieval Agent, receives the customer's data, validates the submitted information, and communicates with connected insurance providers through their APIs. It retrieves the available car insurance offers, including premiums, coverage details, deductibles, and additional benefits, and converts them into a unified format for further processing.

The second agent, the Recommendation Agent, evaluates the retrieved insurance packages using a set of recommendation criteria. These criteria include premium cost, coverage scope, coverage-to-price value, deductible amount, available benefits, suitability for the driver and vehicle profile, provider reliability, and offer availability. The agent ranks all available packages and selects the one with the highest overall suitability score.

The third agent, the Explainer Agent, generates a clear and understandable explanation of the recommendation. It identifies the main factors that contributed to the selection, highlights the strengths of the recommended package, and explains why alternative offers were not chosen. This improves transparency and helps users trust and understand the recommendation process. By combining live offer retrieval, intelligent recommendation logic, and explainable decision-making, THIQA aims to simplify car insurance selection, reduce customer confusion, improve transparency, and enhance the overall digital insurance experience.

CHAPTER THREE: SYSTEM DESIGN AND ARCHITECTURE OF THE THIQA CAR-INSURANCE PLATFORM

3.1 Introduction

This chapter presents the structural and theoretical design of Thiqa, the proposed car-insurance recommendation platform. The chapter focuses exclusively on car (motor) insurance and explains how the platform processes the customer's data, communicates with the insurance companies through their Application Programming Interfaces (APIs — the interfaces that let two software systems exchange data), and delivers an instant, easy-to-understand recommendation to the ordinary customer.

Thiqa is built from two clearly separated components: a User Component used by the ordinary customer, and an Admin Component used by the system administrator. Inside the backend, the platform relies on a three-agent design — a Data Receiver and Retrieval Agent, a Recommendation Agent, and an Explainer Agent. The chapter describes the architecture, the role of each agent and the steps performed inside it, the external API integration mechanism, the system requirements, the UML behavioural models, the database design, and the user-interface flow. Technical terms are explained on first use, and a consolidated glossary is provided in Section 3.8.

3.2 System Architecture

Thiqa is designed as an API-driven, multi-agent recommendation platform that connects the ordinary customer with several car-insurance companies through a centralised backend. The architecture supports real-time data retrieval, automated analysis, and immediate recommendation delivery, so that the customer receives the most suitable car-insurance package in the shortest possible time. Because all offers are pulled live from the insurers, the platform never depends on stored, out-of-date prices.

3.2.1 Architectural Overview

The architecture is organised into three layers. Layer 1 is the presentation layer, which contains the two separated components (the User Component and the Admin Component). Layer 2 is the application backend, which hosts the orchestrator and the three intelligent agents. Layer 3 is the external API layer, which represents the car-insurance companies (such as Tawuniya and Takaful Al-Rajhi) that return live pricing and coverage data. Figure 3.1 shows the overall architecture.

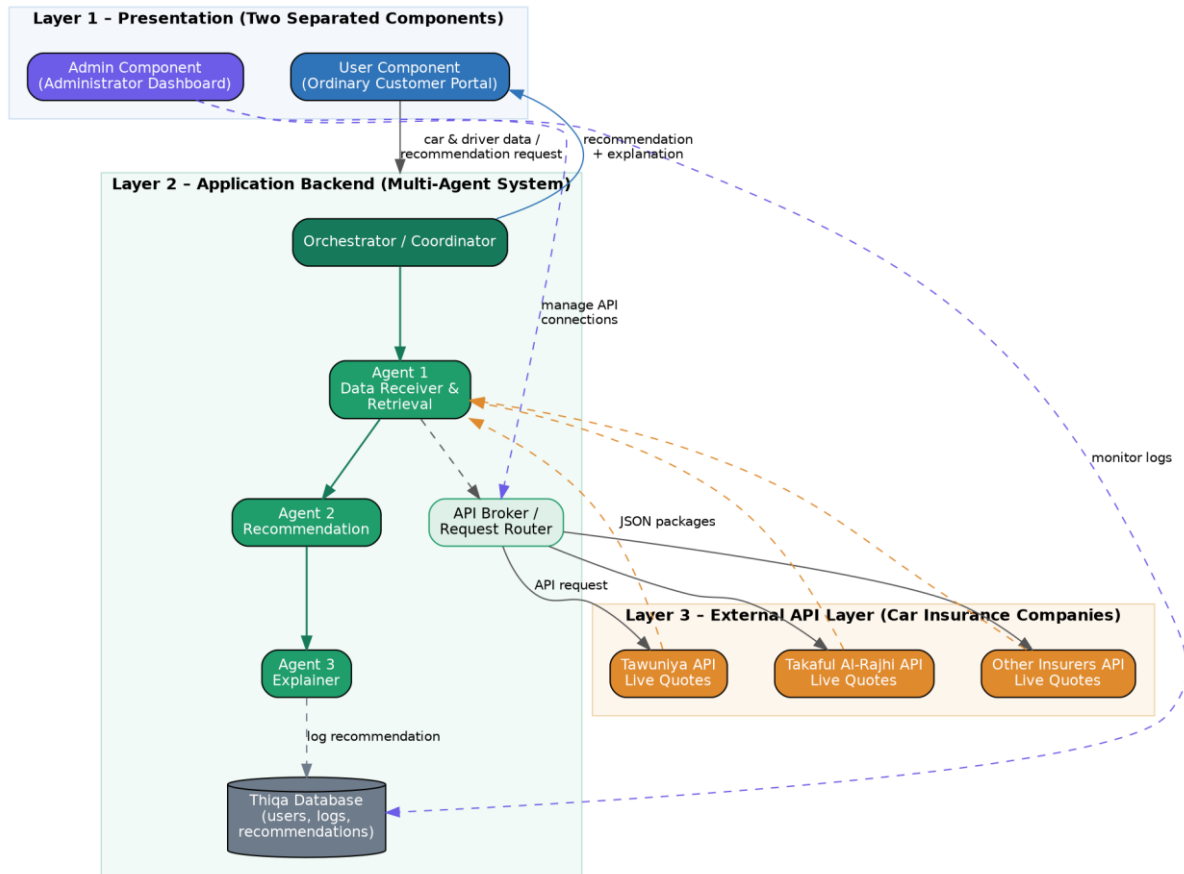


Figure 3.1: Three-Layer Architecture of the Thiqa Car-Insurance Platform

3.2.2 The Two Separated Components

Thiqa deliberately separates the platform into two independent components so that each role works in its own secure space. The User Component is the part the ordinary customer interacts with; it handles registration and login, the entry of car and driver data, and the display of the recommendation and its explanation. The Admin Component is an independent dashboard used only by the administrator to manage the insurance providers, configure and monitor the API connections, and review the logged recommendations. Keeping the two components separate improves security, keeps the customer's journey simple, and lets the administrator supervise the platform without affecting the user experience [35].

3.2.3 The Three-Agent Design

At the core of the backend is a multi-agent system (MAS) — a system made of several cooperating software agents, where each agent is an autonomous component that perceives its input and acts to achieve a goal [23], [24]. Thiqa uses three agents arranged as a one-directional pipeline: the Data Receiver and Retrieval Agent (Agent 1) collects the customer's data and the live insurer offers; the Recommendation Agent (Agent 2) selects the best car-insurance package; and the Explainer Agent (Agent 3) explains why that package was chosen. This separation of tasks keeps each agent simple, makes the platform easier to maintain and extend, and allows the work to be reasoned about one stage at a time. Figure 3.2 shows the end-to-end pipeline.

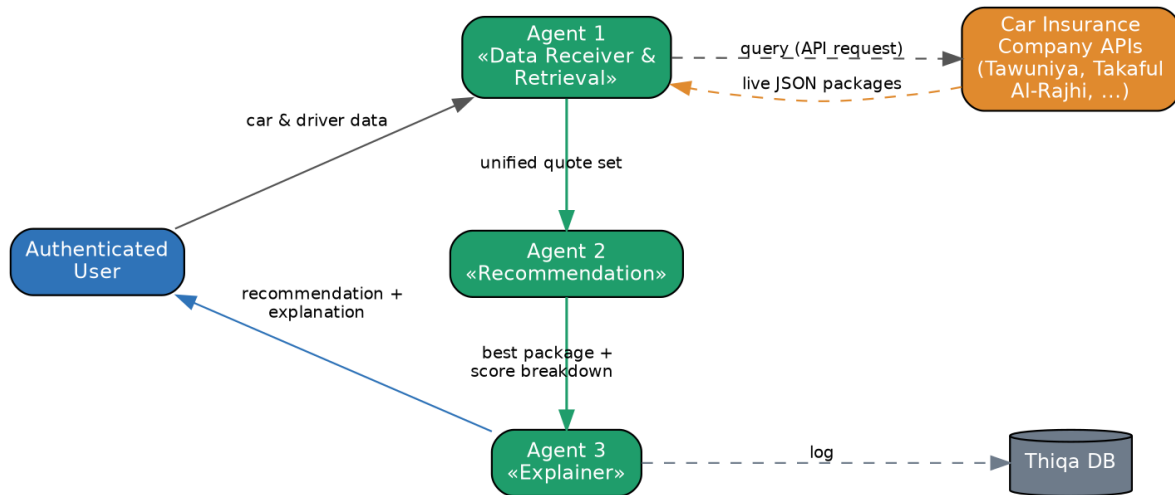


Figure 3.2: The Three-Agent Pipeline (Receiver → Recommendation → Explainer)

3.2.3.1 Agent 1 – Data Receiver and Retrieval Agent

Agent 1 is the entry point of the backend. After the customer logs in and submits the car and driver data, this agent receives that data, sends it to each connected car-insurance company through its API, and then receives back all of the available car-insurance packages (premiums, coverage types, deductibles, and add-ons) from companies such as Tawuniya, Takaful Al-Rajhi, and others. Agent 1 makes no decision; it only collects, validates, and standardises the offers into a single unified quote set that it passes to Agent 2. Figure 3.3 shows the agent in its context, and Figure 3.4 details the steps performed inside it.

Figure: Agent 1 Context Diagram

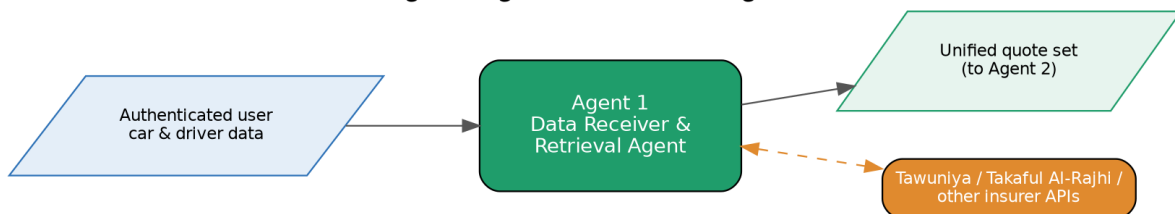
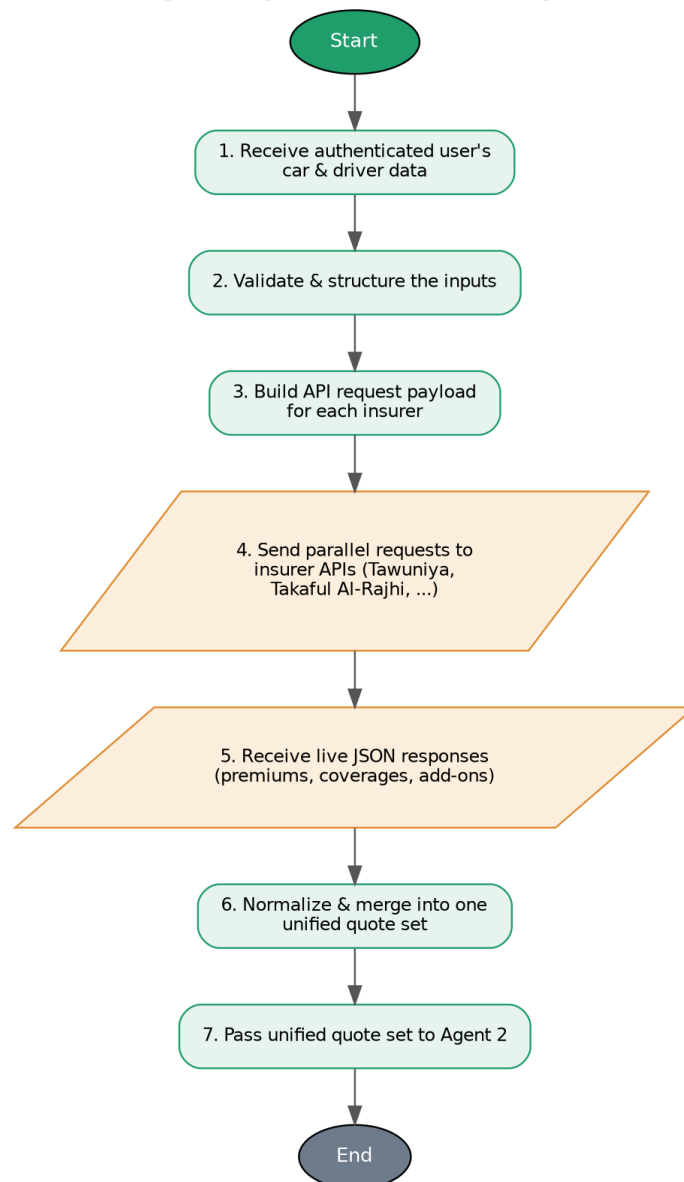


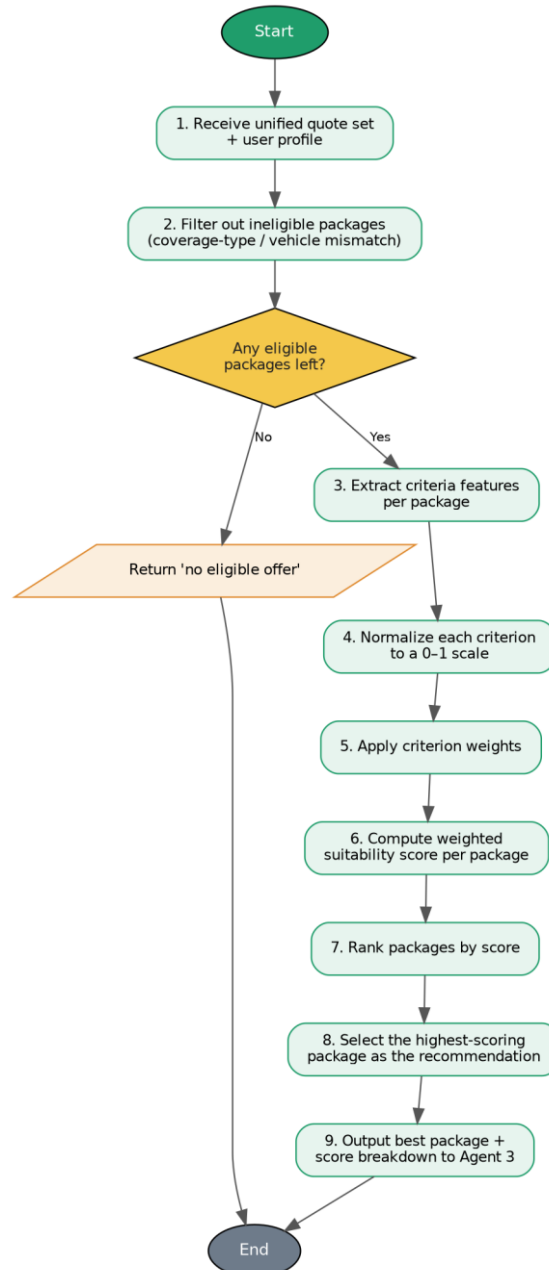
Figure 3.3: Agent 1 – Context Diagram

Internally, Agent 1 performs the following steps: it receives the authenticated customer's data, validates and structures the inputs, builds an API request payload for each insurer, sends the requests to the insurer APIs (in parallel, to save time), receives the live JSON responses, normalises and merges them into one unified quote set, and finally passes that set to the Recommendation Agent.

Figure: Agent 1 - Internal Steps*Figure 3.4: Agent 1 – Internal Steps*

3.2.3.2 Agent 2 – Recommendation Agent

Agent 2 is the decision-making agent. Its goal is to select the single best car-insurance package for the customer based on that customer's data. It receives the unified quote set from Agent 1 together with the customer's profile (driver and vehicle details), evaluates every package against a defined set of criteria, and outputs the highest-scoring package together with a full score breakdown for Agent 3. Figure 3.5 shows the agent in context, and Figure 3.6 details its internal steps.

Figure: Agent 2 Context Diagram*Figure 3.5: Agent 2 – Context Diagram***Figure: Agent 2 - Internal Steps (Recommendation Logic)***Figure 3.6: Agent 2 – Internal Steps (Recommendation Logic)*

How the agent works. Recommendation in Thiqa follows the standard recommender-system pipeline of filtering candidates and then ranking them before a final choice is made [28]. Agent 2 works in the following way. (1) It receives the unified quote set and the customer profile. (2) It

filters out any package that the customer is not eligible for or that does not match the request — for example, a package whose coverage type does not match the requested one, or that does not apply to the customer’s vehicle. (3) For each remaining package it extracts a set of comparable features (the criteria). (4) It normalises each criterion to a common 0–1 scale so that values measured in different units (riyals, yes/no benefits, ratings) can be compared fairly. (5) It multiplies each normalised criterion by a weight that reflects how important that criterion is. (6) It adds the weighted values to produce one suitability score for each package, computed as $S = \sum (w_i \times x_i)$, where x_i is the normalised value of criterion i and w_i is its weight, and the weights add up to 1. (7) It ranks the packages by their score. (8) It selects the package with the highest score as the recommendation, and (9) outputs it with its score breakdown to Agent 3.

The criteria used to score each package. Each package is scored on eight criteria. Premium and coverage carry the largest weights:

- **1. Premium (price):** the annual cost. A lower premium raises the score, weighed against coverage.
- **2. Coverage type and scope:** Comprehensive or Third-Party Liability (TPL), and the range of risks covered (collision, theft, fire, natural events). Broader, more relevant coverage raises the score.
- **3. Coverage-to-price value:** the ratio of protection provided to cost. Prevents both very cheap but weak and expensive but unnecessary packages from winning.
- **4. Deductible:** the amount the customer pays per claim. A lower deductible raises the score.
- **5. Add-ons and benefits:** extras such as agency repair, roadside assistance, replacement car, windshield cover, and geographic extension. Relevant add-ons raise the score.
- **6. Fit to the driver and vehicle profile:** how well the package suits driver age, vehicle value, vehicle age, and purpose of use.
- **7. Provider reliability:** the insurer’s reputation and claim-settlement record, used as a trust signal and tie-breaker.
- **8. Availability and freshness:** only packages retrieved live in the current session are eligible.

Why the agent chose one package over the others. The agent does not simply pick the cheapest offer; it picks the package with the highest weighted suitability score. In the real worked example shown later in the platform screenshots (Figure 3.16), a Third-Party-Liability package from Tawuniya at SAR 483 per year is recommended even though two other offers are cheaper — Medgulf at SAR 469 and ACIG at SAR 428 — and one is more expensive, Takaful Al-Rajhi at SAR 507. Tawuniya wins because it achieves the best overall balance for this customer, whose stated priority was the most affordable yet most suitable option: it pairs a competitive premium with the widest accredited workshop network in the Kingdom, fast nationwide claim settlement, and high provider reliability, which together give it the best coverage-to-price value and the highest overall match with the customer’s preferences. The cheaper ACIG and Medgulf offers score lower on reliability and service quality, and the more

expensive Takaful Al-Rajhi offer scores lower on the value criterion. The chosen package, its score, and the criterion-by-criterion breakdown are then passed to Agent 3.

3.2.3.3 Agent 3 – Explainer Agent

Agent 3 makes the decision understandable. Its goal is to explain, in plain language, why the chosen car-insurance package was recommended to the customer, including the reasons and the criteria behind the choice. Providing such an explanation is the field known as explainable AI (XAI — techniques that make an automated decision understandable to a human), and it matters because an ordinary, non-technical customer must understand why an option was selected in order to trust it [29], [30]. Figure 3.7 shows the agent in context and Figure 3.8 details its steps.

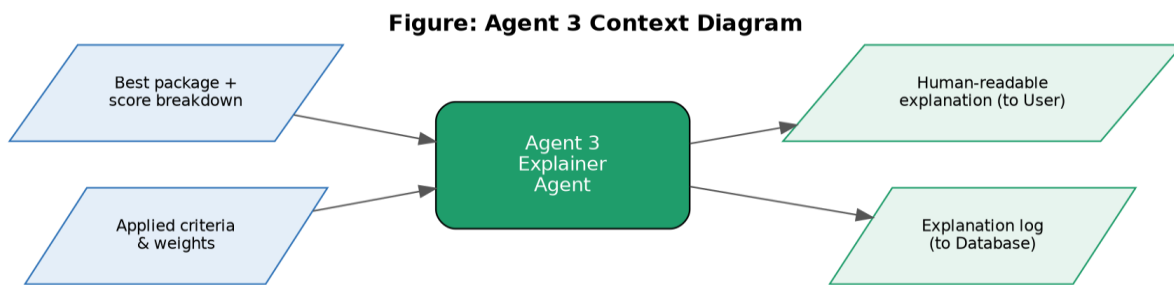
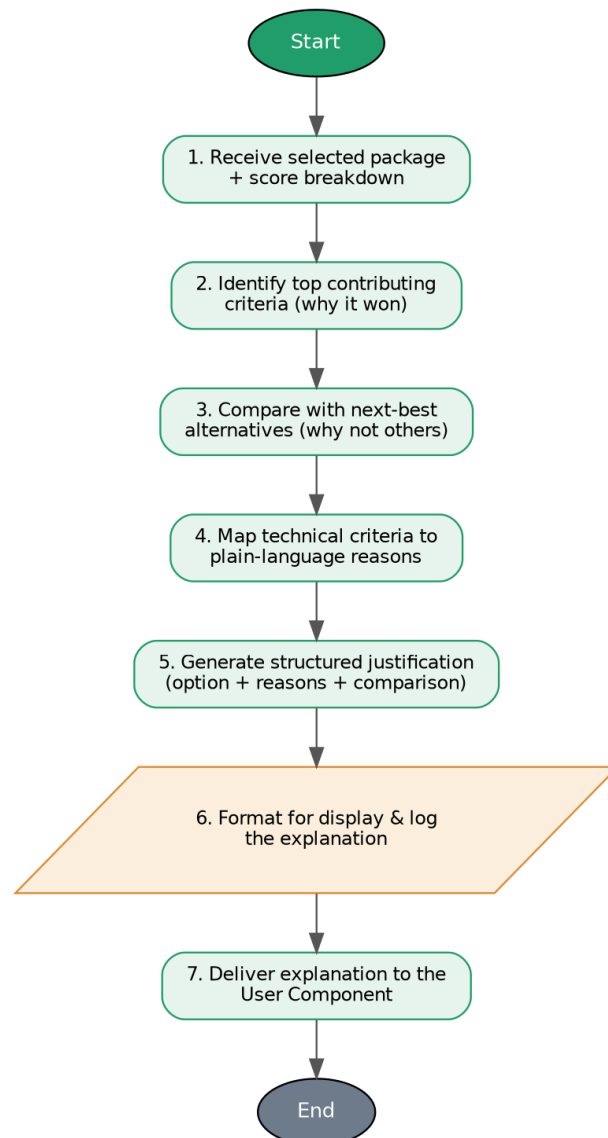


Figure 3.7: Agent 3 – Context Diagram

Internally, Agent 3 receives the selected package and its score breakdown, identifies the criteria that contributed most to the choice (why this package won), compares the choice against the next-best alternatives (why the others were not chosen), maps the technical criteria to plain-language reasons, generates a structured justification (the recommended option, the main reasons, and a short comparison), formats it for display, logs it to the database, and delivers it to the User Component.

Figure: Agent 3 - Internal Steps (Explanation)*Figure 3.8: Agent 3 – Internal Steps*

3.3 External API Integration Mechanism

The external API integration mechanism is the process that lets Thiqa obtain live car-insurance data from the insurers. Rather than storing fixed offers, the platform sends a request to each insurer's API whenever the customer asks for a recommendation [25], [26].

Data ingestion. When the customer requests a recommendation, Agent 1 builds an API request carrying the car and driver data and sends it to each insurer's endpoint (a specific URL of the API). Each insurer replies in real time with its current packages. Because the requests are sent in parallel, the offers from several companies arrive within a few seconds.

Data format. The insurers return their data as JSON (JavaScript Object Notation — a lightweight, text-based data format) [27]. Agent 1 parses each JSON response and converts it

into one unified structure so that packages from different companies can be compared consistently and objectively by Agent 2.

3.4 System Requirements

The platform must satisfy a set of functional and non-functional requirements that define what it should do and how well it should perform [35].

3.4.1 Functional Requirements

- **User Authentication:** The system shall let the customer register and log in before entering data, and shall keep the User and Admin components separate.
- **Car & Driver Data Entry:** The system shall let the customer enter the same car and driver data that insurance companies require (e.g. vehicle serial number, make/model/year, estimated value, driver date of birth, purpose of use, and requested coverage type).
- **Live Offer Retrieval:** The system (Agent 1) shall send the customer's data to the insurer APIs and retrieve all live car-insurance packages.
- **Recommendation:** The system (Agent 2) shall select the best car-insurance package by scoring the live packages against the defined criteria.
- **Explanation:** The system (Agent 3) shall present a clear explanation of why the chosen package was recommended.
- **Result Display:** The system shall display the recommendation and its explanation in a simple format suitable for a non-technical customer.
- **Admin Control:** The system shall let the administrator manage insurance providers and API connections and review recommendation logs.
- **Logging:** The system shall store each recommendation and explanation for auditing and future improvement.

3.4.2 Non-Functional Requirements

- **Response Time:** Because the platform fetches data from external APIs and processes it through three agents, the whole recommendation shall be produced within a few seconds to keep the experience smooth for the ordinary customer.
- **Reliability:** The system shall continue to work even if one insurer API is slow or unavailable.
- **Scalability:** The system shall support more customers and more insurers without major redesign.
- **Usability:** The interface shall be simple and intuitive for users with no technical background.
- **Security:** The system shall protect the customer's data and the API credentials.

- **Data Accuracy:** The recommendation shall rely on live data retrieved from the insurer APIs.
- **Flexibility:** The system shall adapt to changes in the insurers' APIs over time.

3.5 System Modeling and UML Diagrams

This section models the behaviour of Thiqa using the Unified Modeling Language (UML — a standard notation for modelling software systems) [31], [32], [33]. The models make the design easier to understand before implementation.

3.5.1 Use Case Diagram

The use case diagram identifies the actors and the services offered to each one [34]. The two actors are the Ordinary User and the Administrator, and the diagram reflects the two separated components. The user can register/login, enter car and driver data, request a recommendation, and view the recommendation and its explanation; the administrator can log in, manage providers and API connections, monitor activity, and view the recommendation logs. Figure 3.9 shows the diagram.

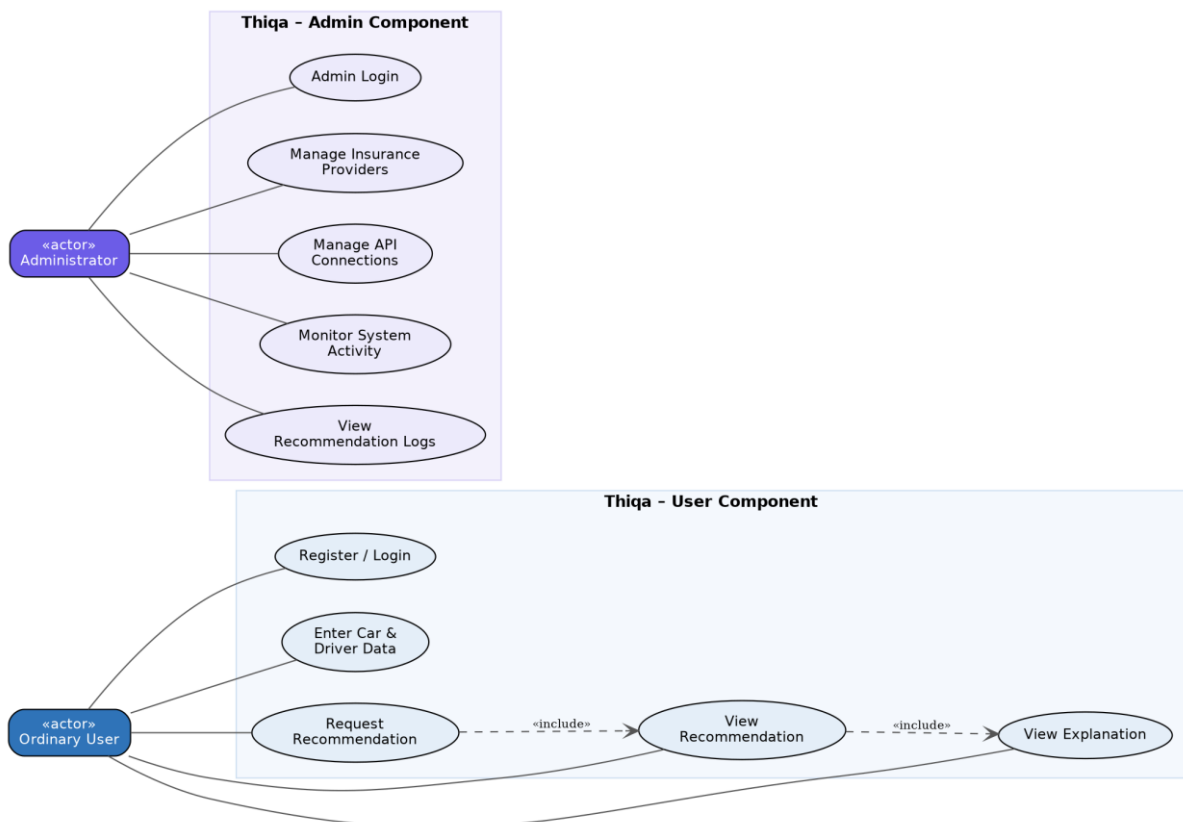


Figure 3.9: Use Case Diagram of the Thiqa Platform (Two Separated Components)

3.5.2 Sequence Diagram

The sequence diagram is the most important behavioural model in this chapter: it shows the time-ordered messages exchanged when the customer requests a recommendation. The customer logs in and submits the car and driver data; the system forwards the data to Agent 1, which queries the insurer APIs and receives live JSON packages; Agent 2 filters, scores, and selects the best package; Agent 3 builds the explanation; the result is logged and finally displayed to the customer. Figure 3.10 shows the diagram.

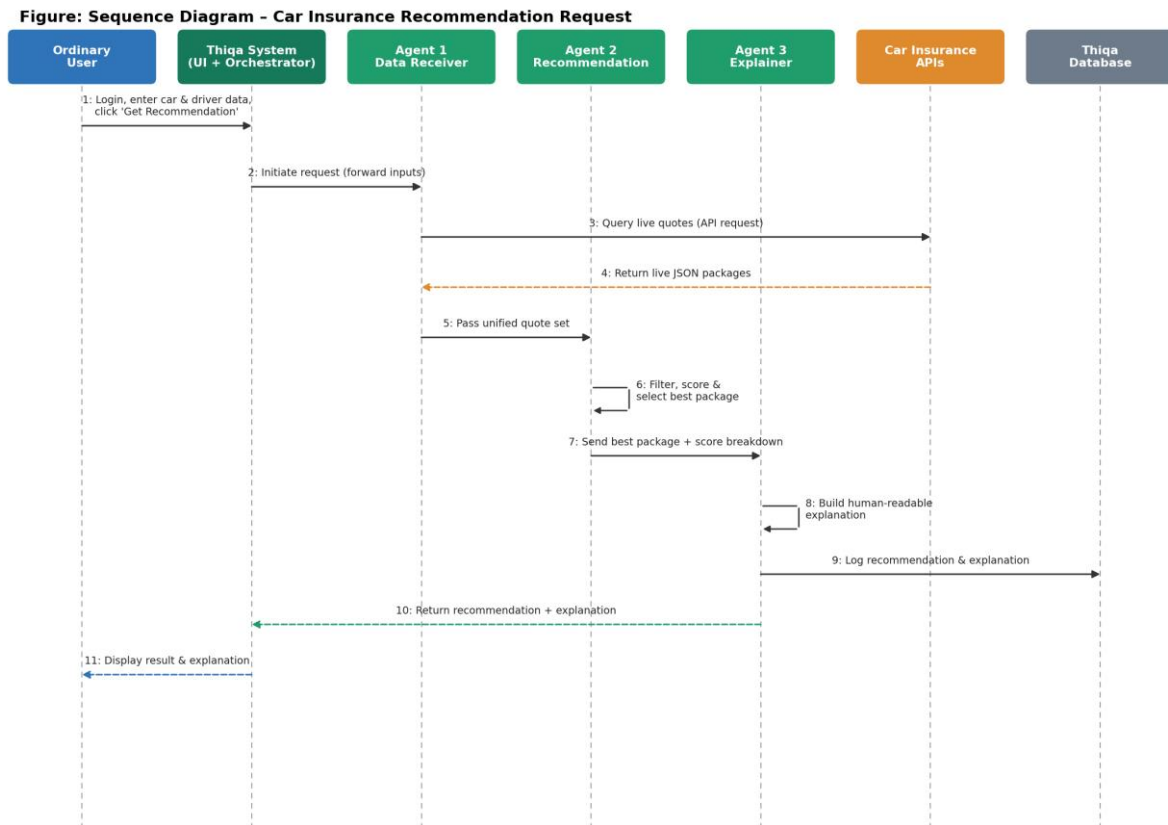
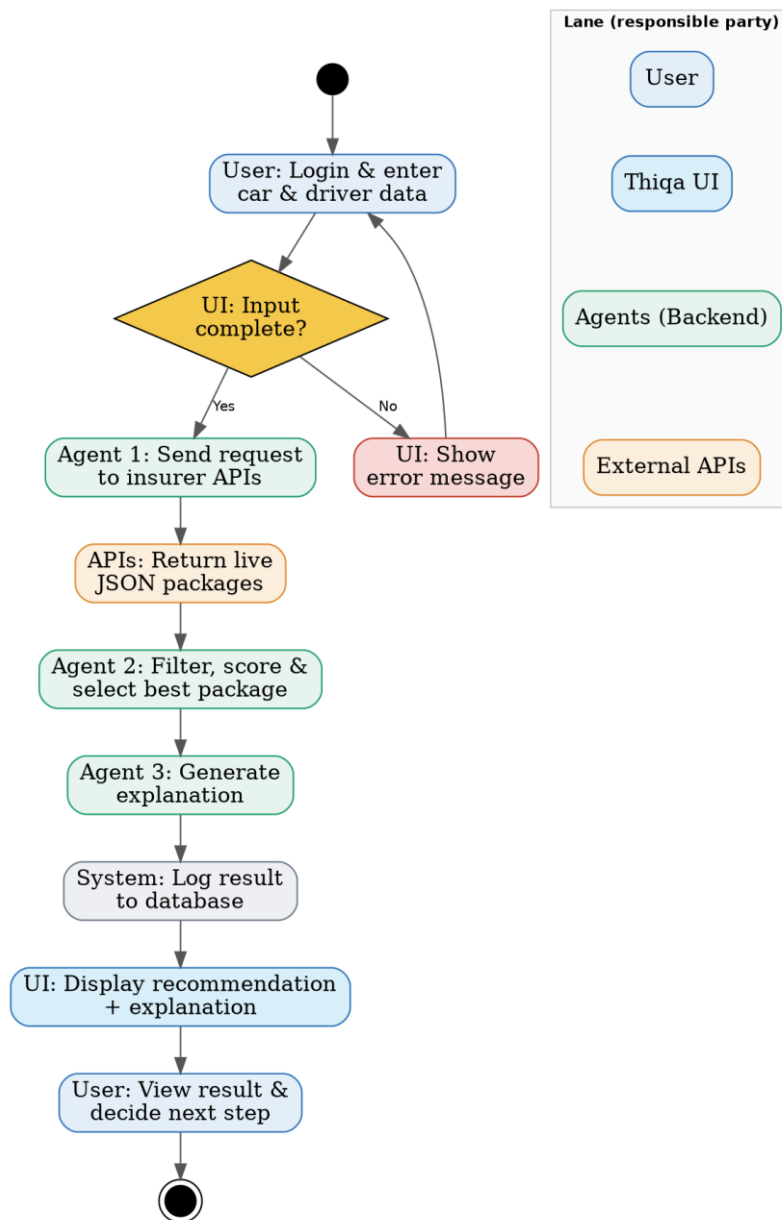


Figure 3.10: Sequence Diagram for the Car-Insurance Recommendation Request

3.5.3 Activity Diagram

The activity diagram shows the workflow from start to finish and which party is responsible for each action. It begins when the customer logs in and enters data, the system validates the input, Agent 1 retrieves the live offers, Agent 2 selects the best package, Agent 3 generates the explanation, the result is logged and displayed, and the process ends. Figure 3.11 shows the diagram.

Figure: Activity Diagram - Recommendation Process*Figure 3.11: Activity Diagram for the Recommendation Process*

3.6 Database Design

Because all car-insurance offers are retrieved live from the insurer APIs, the database does not store fixed company offers. Instead, it stores the internal data needed to run the platform: users, customer profiles, vehicles, the requests, the live packages that were retrieved for each request, the chosen recommendation, and its explanation. This keeps a complete, traceable log of what was shown to each customer [36].

3.6.1 Entity Relationship Diagram (ERD)

The ERD models the database tables (entities) and the relationships between them. Each table has a primary key (PK — a field that uniquely identifies a row) and may have foreign keys (FK — fields that link to another table). A user owns one or more customer profiles; a profile owns one or more vehicles and submits one or more quote requests; each request retrieves many live packages from many providers; each request produces one recommendation, which references the selected package and has one explanation. Figure 3.12 shows the ERD.

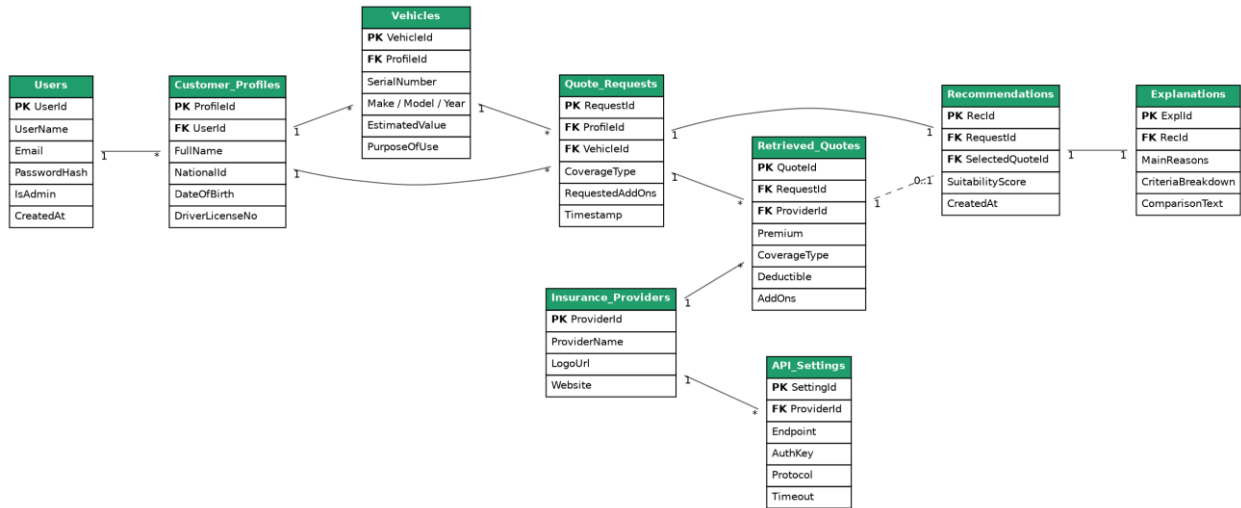


Figure 3.12: Entity Relationship Diagram for the Thiqa Platform (Car Insurance)

3.6.2 Data Dictionary

The data dictionary documents each table, its fields, their data types, and their purpose. Table 3.1 summarises the main tables of the platform.

Field	Data Type	Description
Users		
UserId	INTEGER (PK)	Unique identifier for each account
UserName	VARCHAR(50)	Login name
Email	VARCHAR(100)	Contact email
PasswordHash	VARCHAR(255)	Securely hashed password
IsAdmin	BOOLEAN	Distinguishes admin from ordinary user
CreatedAt	DATETIME	Account creation timestamp
Customer_Profiles		
ProfileId	INTEGER (PK)	Unique profile identifier
UserId	INTEGER (FK)	Owning user account
FullName	VARCHAR(100)	Customer full name
NationalId	VARCHAR(20)	National ID / Iqama number
DateOfBirth	DATE	Used to derive driver age
DriverLicenseNo	VARCHAR(30)	Driving licence number
Vehicles		

VehicleId	INTEGER (PK)	Unique vehicle identifier
ProfileId	INTEGER (FK)	Owning customer profile
SerialNumber	VARCHAR(40)	Vehicle sequence / chassis number
MakeModelYear	VARCHAR(80)	Make, model and year
EstimatedValue	DECIMAL(10,2)	Market value of the car
PurposeOfUse	VARCHAR(40)	Personal / commercial use
Quote_Requests		
RequestId	INTEGER (PK)	Unique recommendation request
ProfileId	INTEGER (FK)	Requesting profile
VehicleId	INTEGER (FK)	Vehicle being insured
CoverageType	VARCHAR(30)	Requested: TPL or Comprehensive
RequestedAddOns	TEXT	Optional add-ons requested
Timestamp	DATETIME	Time of the request
Insurance Providers		
ProviderId	INTEGER (PK)	Unique insurer identifier
ProviderName	VARCHAR(100)	e.g. Tawuniya, Takaful Al-Rajhi
LogoUrl	VARCHAR(255)	URL of the insurer logo
Website	VARCHAR(255)	Insurer website
API_Settings		
SettingId	INTEGER (PK)	Unique API configuration
ProviderId	INTEGER (FK)	Related insurer
Endpoint	VARCHAR(255)	Base URL of the insurer API
AuthKey	VARCHAR(255)	Authentication key / token
Protocol	VARCHAR(40)	e.g. REST / JSON
Timeout	INTEGER	Max wait time per request (seconds)
Retrieved_Quotes		
QuoteId	INTEGER (PK)	One package returned by an insurer
RequestId	INTEGER (FK)	Parent request
ProviderId	INTEGER (FK)	Source insurer
Premium	DECIMAL(10,2)	Annual premium of the package
CoverageType	VARCHAR(30)	TPL / Comprehensive
Deductible	DECIMAL(10,2)	Customer out-of-pocket per claim
AddOns	TEXT	Benefits included in the package
Recommendations		
RecId	INTEGER (PK)	One recommendation per request
RequestId	INTEGER (FK)	Parent request
SelectedQuoteId	INTEGER (FK)	The chosen package
SuitabilityScore	DECIMAL(5,4)	Weighted score of the chosen package
CreatedAt	DATETIME	When it was generated
Explanations		

ExplId	INTEGER (PK)	One explanation per recommendation
RecId	INTEGER (FK)	Parent recommendation
MainReasons	TEXT	Top reasons in plain language
CriteriaBreakdown	TEXT	Per-criterion contribution
ComparisonText	TEXT	Why the alternatives were not chosen

Table 3.1: Data Dictionary for the Thiqa Platform

3.7 User Interface and Navigation Flow

The user-interface design keeps the customer's experience simple, clear, and fast, while keeping the administrator's tools in a separate, secure area.

3.7.1 Navigation Flow

The navigation flow is kept short and direct so the customer reaches the result with the fewest possible clicks. The customer moves from the landing page to login, then to the car and driver data form, then through processing by the three agents, and finally to the recommendation-and-explanation page, from which a new inquiry can be started. The administrator follows a completely separate path: admin login, then a dashboard from which providers, API connections, and activity logs are managed. Figure 3.13 shows both flows.

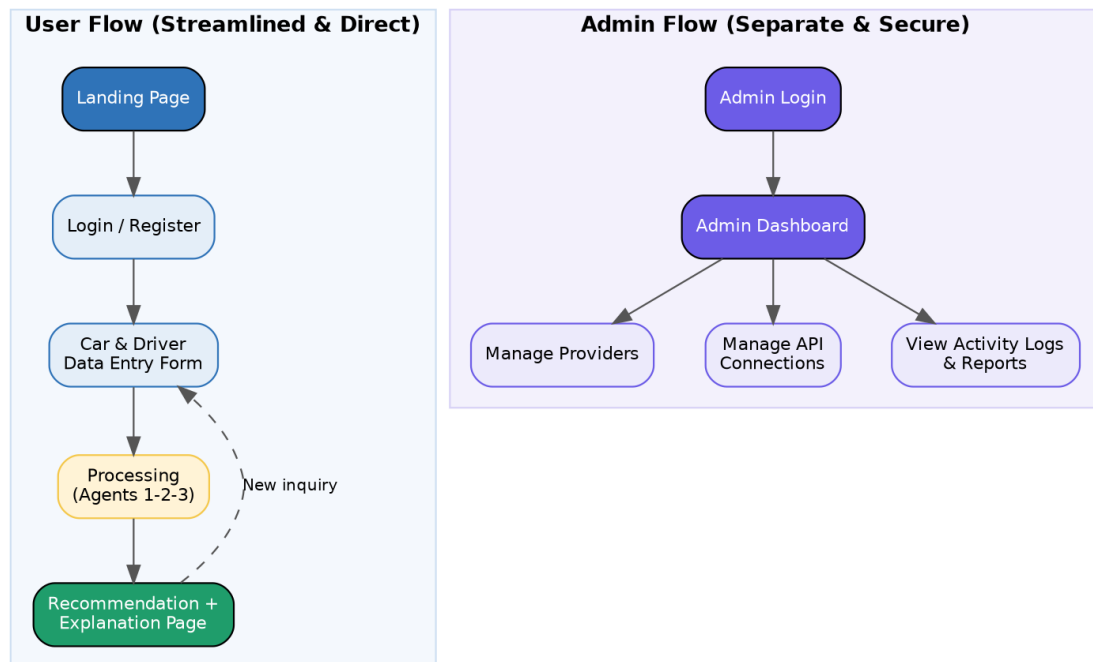


Figure 3.13: Navigation Flow for the User and Admin Components

3.7.2 User Interface Screens

The following figures are screenshots of the implemented Thiqa interface. Because the platform serves customers in Saudi Arabia, the interface itself is presented in Arabic; each screen is

described here in English. Three screens make up the complete customer journey: the landing page, the data-entry page, and the recommendation-and-explanation page.

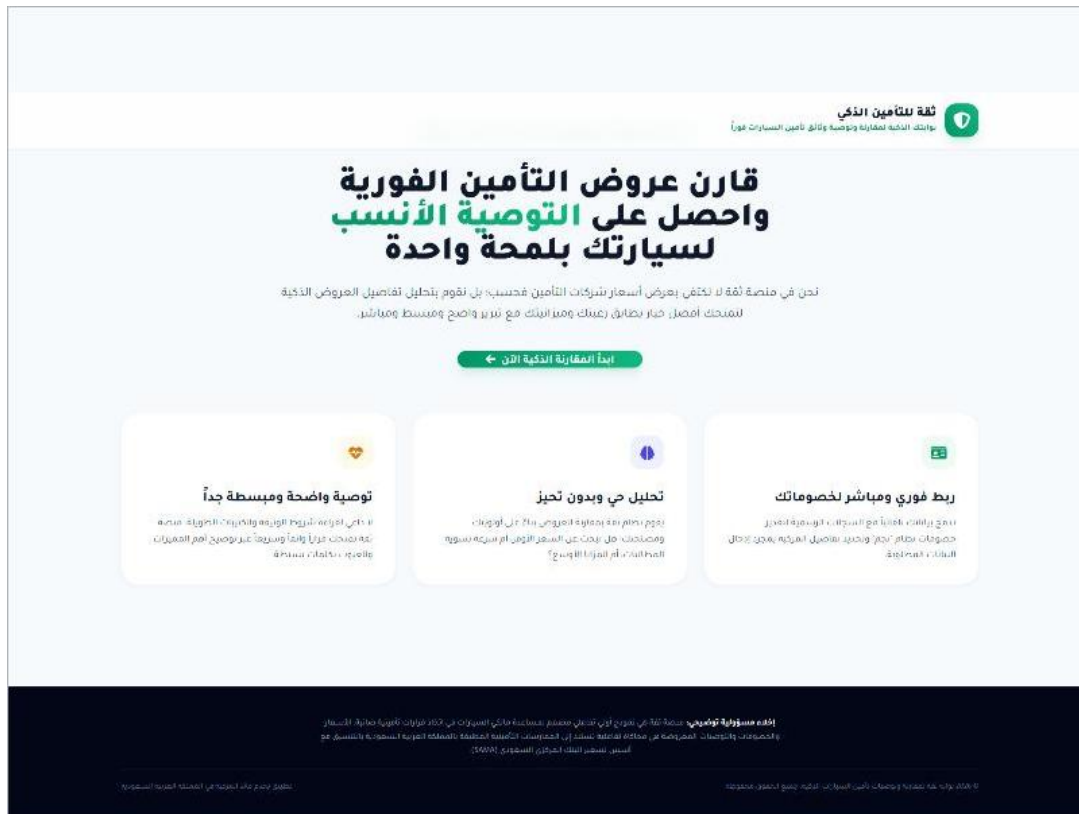


Figure 3.14: Thiqa Landing Page (User Component)

The landing page introduces the platform and leads the customer to a single call-to-action that starts the smart comparison. It states the value of the service in one line — compare live car-insurance offers and receive the most suitable recommendation at a glance — and summarises the three promises of the system: a clear and simple recommendation, neutral analysis, and a direct connection to the insurers. It carries no technical detail, so the ordinary customer is guided straight into the workflow.

Figure 3.15: Car and Driver Data-Entry Page (User Component)

The data-entry page collects exactly the same fields that car-insurance companies require to price a policy: the national ID or Iqama number, the date of birth, the vehicle serial (chassis) number, the manufacturing year, the primary purpose of use and annual mileage, the preferred coverage type — Comprehensive or Third-Party Liability (TPL) — and the customer's main priority, such as obtaining the most affordable and most suitable price. This single form is the input that Agent 1 sends to the insurers' APIs. The interface is deliberately kept to one short form so the process stays fast and easy.

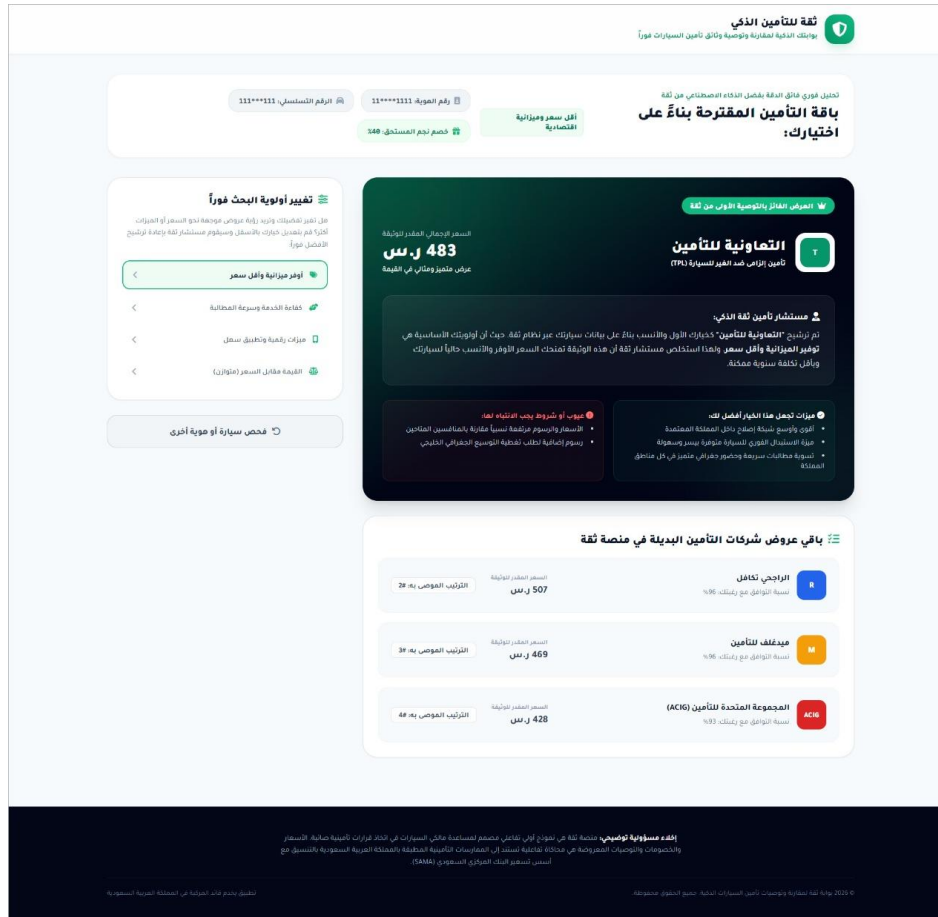


Figure 3.16: Recommendation and Explanation Page (User Component)

The recommendation page presents the output of the three agents. At the top it shows the winning package chosen by Agent 2 — here a Tawuniya Third-Party-Liability policy at SAR 483, marked as the top recommendation and labelled an excellent value. Directly beneath it, the “Thiqa smart advisor” block is the output of Agent 3 (the Explainer): in plain language it states why this package was selected for the customer, then lists the advantages that make it the best choice (a wide accredited workshop network, instant live retrieval, and fast nationwide claims) together with the trade-offs to be aware of (a premium that is somewhat higher than some competitors, and extra fees for Gulf geographic coverage). Below the recommendation, the remaining offers retrieved in the same session are listed and ranked — Takaful Al-Rajhi at SAR 507, Medgulf at SAR 469, and ACIG at SAR 428 — each shown with its match percentage against the customer’s preferences, so the customer can see not only what was recommended but how it compares with the alternatives.

3.8 Chapter Summary

This chapter presented the design of Thiqa, a car-insurance recommendation platform built from two separated components (a User Component and an Admin Component) and a backend multi-agent system of three agents. Agent 1 receives the customer’s data after login and retrieves all live packages from the insurers’ APIs (such as Tawuniya and Takaful Al-Rajhi); Agent 2 selects

the best package by scoring the live offers against a set of weighted car-insurance criteria; and Agent 3 explains, in plain language, why that package was chosen.

The chapter detailed the role and internal steps of each agent, explained in full the criteria and scoring logic of the Recommendation Agent and the reasoning of the Explainer Agent, and modelled the system with use case, sequence, and activity diagrams, an ERD with a data dictionary, and the navigation flow together with screenshots of the implemented user interface. Together, these designs show that Thiqa is simple for the customer, organised in structure, and suitable for fast, dynamic, and explainable car-insurance recommendation.

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